

W9 Exercises



Discuss with TAs or consult the TA session notes if you run into any difficulties!

Task 0

- Download and import the dataset `restaurants`

Task 1: Nohara Hiroshi: Hiru Meshi no Ryuugi (野原廣智 午餐的流派)

- This practice is fully credited to last year's TA, 康柏賢. Enjoy!
 - For those who don't know Nohara Hiroshi, he is an anime character (see descriptions below)



"My name is Hiroshi Nohara (野原廣志). I'm an office worker and often eat lunch out. Based on the state of my body and my wallet, I quickly decide what to eat each day. You could say that while I'm a business expert, I'm also a lunch expert."

- Hiroshi keeps a list of restaurants in the dataset `restaurants`. The file has five columns:
 - `name`: The restaurant's name
 - `type`: The type of cuisine
 - `closed`: The restaurant's closed day; **some restaurants are permanently closed**
 - `lunch_set`: Whether the restaurant offers lunch sets at noon
 - `rating`: Hiroshi's personal rating for the restaurant



Today is **Thursday**. Hiroshi feels like he would like some seafood, but curry or noodles would also be fine. Since he had fast food yesterday, he'd rather avoid it today. It's close to the end of the month, so it would be better if the restaurant offers lunch sets.

- Follow the instructions below to help Hiroshi choose the restaurants that best fit his needs: (**Hint:** make good use of `filter()` , `select()` , `mutate()` , `arrange()` , `%>%` in following steps)
 - Step 0: Import `dplyr`
 - Step 1: Filter the restaurants that are open today
 - Step 2: Add a column `lunch_set_score` that is `1.5` if the restaurant offers lunch sets and `0` otherwise
 - Step 3: Add a column `preference_score` that takes value `1.5` for `seafood` , `1` for `curry` and `noodle` , `-2` for `fastfood` , and `0` otherwise
 - Step 4: Add a column `total_score` that sums up the columns `rating` , `lunch_set_score` and `preference_score`
 - Step 5: Remove the `lunch_set_score` and `preference_score` columns
 - Step 6: Arrange the rows in descending order of `total_score`
 - Step 7: Print the top 5 rows with the highest `total_score`